ISCRAM 2026
Building Stronger Futures

Voices Across Crises: Stakeholder Perspectives and Opportunities for AI Speech Technologies

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Theme: **Building Stronger Futures: Ensuring Public Safety in Times of Crisis**

TITLE

Voices Across Crises: Stakeholder Perspectives and Opportunities for AI Speech Technologies

Universiteit Leiden
The Netherlands

NIPV Nederlands Instituut Publieke Veiligheid

ABSTRACT

Effective communication is critical for public safety during crises, particularly for vulnerable communities facing barriers to accessing timely information. Emergency responders, humanitarian organizations, and affected populations rely on rapid information exchange across several key functions: situational awareness, information management, needs assessment and coordination, risk communication and early warning, and recovery and long-term monitoring. Crisis communication often happens through spoken channels such as emergency hotlines, radio, and voice messaging apps. These channels are inherently multilingual and speech-based, creating significant barriers for vulnerable communities and responders alike. Actors across the crisis ecosystem, government agencies, humanitarian NGOs, broadcasters, community leaders, and affected populations, must rapidly exchange information across languages under pressure.

Recent advances in artificial intelligence (AI), including automatic speech recognition (ASR), speech translation, and multilingual large language models (LLMs), have opened new opportunities for crisis communication. However, these technologies are often developed without a thorough understanding of operational realities. Different stakeholders have distinct needs: government agencies coordinate logistics, NGOs manage field operations, and community leaders relay local needs. It remains unclear whether emerging speech and voice technologies adequately address the needs of responders, vulnerable communities, and other crisis actors.

This workshop explores how voice technologies and AI tools could support communication and information sharing during emergencies, with a focus on vulnerable and linguistically diverse communities. We draw on ongoing surveys and interviews with crisis communication stakeholders, including government representatives, radio and streaming organizations, humanitarian NGOs, and community leaders, to ground the discussion in real-world experiences. Participants will reflect on these findings, share their own perspectives, and identify key communication challenges. Small group activities will map communication practices across the crisis ecosystem and explore potential applications of speech and voice technologies across use cases such as situational awareness, early warning, and needs coordination, including associated risks and limitations.

This workshop is intended for crisis management researchers, practitioners, policymakers and technology researchers. Attendees will develop a clearer understanding of multilingual communication challenges in crisis settings and the potential of AI and voice tools to address them. Expected outcomes include a validated understanding of speech-based communication practices across the crisis ecosystem, prioritized use cases for speech and AI technologies, and concrete insights to inform future research.

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PRESENTER/DEMONSTRATOR

*Belu Ticona**

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George Mason University

Belu Ticona is a Ph.D. student in Computer Science at George Mason University, researching human-centered speech technologies for crisis scenarios, with a focus on voice technologies that address communication barriers in high-risk and disaster-affected communities. She has experience organizing academic activities in this space, including Birds of a Feather sessions on Language Technologies for Crisis Preparedness and Response (LT4CPR) at major NLP venues such as ACL and COLING, and is co-organizing the upcoming LT4CPR workshop at AACL 2026.

She is also conducting research in partnership with CLEAR Global, a nonprofit focused on language access and communication for underserved communities, leading the development of speech technology infrastructure grounded in the needs and challenges of crisis stakeholders

This will be her first time attending ISCRAM, and she plans to be present at the conference.

Antonis Anastasopoulos

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George Mason University

Antonios Anastasopoulos is an Assistant Professor in the Computer Science Department at George Mason University and a Collaborating Senior Researcher at Archimedes AI Research Center. He received his Ph.D. in Computer Science from the University of Notre Dame and previously held a postdoctoral position at Carnegie Mellon University.

His research focuses on multilingual natural language processing, with an emphasis on machine translation and speech recognition for low-resource and endangered languages, as well as cross-lingual transfer and model robustness. His work has strong relevance to crisis contexts, including his involvement in the Translation Initiative for COVID-19 (TICO-19), a large-scale collaboration between academia and industry to support multilingual communication during the COVID-19 pandemic. He is also a recipient of a NSF research grant supporting work on LT4CPR.

Dr. Anastasopoulos has extensive experience organizing and leading academic events. He has served in multiple roles at major NLP venues such as Association for Computational Linguistics conferences and has co-organized more than five ACL-affiliated workshops. He is also a co-organizer of the upcoming LT4CPR workshop at AACL 2026.

William Lewis*

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University of Washington

William D. Lewis is a researcher with extensive experience in machine translation (MT), low-resource languages, and language technologies for crisis response. For nearly 14 years, until December 2020, he worked on the Machine Translation Incubation team at Microsoft, most of that time embedded at Microsoft Research and later Microsoft Azure. Prior to that, he helped found and teach in the Computational Linguistics M.S. Program at the University of Washington with Emily Bender and Fei Xia. Earlier, he served as faculty at California State University, Fresno, where he helped establish the university's Computational Linguistics and Cognitive Science programs and served as the inaugural chair of both.

His research focuses on MT for low-resource and endangered languages, data curation, and the use of language technologies in crisis settings. At Microsoft Translator, he led efforts to develop and deploy MT for under-resourced languages including Haitian Kreyòl, White Hmong, Querétaro Otomí,, Icelandic, Canadian French, and Inuktitut among others. Following the 2010 Haiti earthquake, he helped lead the rapid development of the world's first commercial Haitian Kreyòl MT system in fewer than five days, an achievement that helped shape his long-term research agenda on crisis response. This work later informed broader efforts such as TICO-19 and the Language Technologies for Crisis Preparedness and Response (LT4CPR) project, whose resources were used by Translators without Borders, now part of CLEAR Global.

Dr. Lewis is also a co-PI on the NSF-funded LT4CPR project in collaboration with colleagues at the University of Washington and George Mason University. He brings substantial experience in research, academic program building, and graduate training in computational linguistics and crisis-oriented language technologies.

*Corresponding presenter

MODE OF WORKSHOP / TUTORIAL

The workshop requires a room with seating arrangements that support small group discussions. A projector and screen are needed for the presentation of survey and interview findings. A whiteboard or flip charts would be helpful for the group activities. No technical setup or software installation is required from participants.